Case Report

Zimmermann–Laband syndrome-associated hereditary gingival fibromatosis

Jaishree Tukaram Kshirsagar, Kalaiselvan Dharani, Priyangha Thangavel

Abstract:

Hereditary gingival fibromatosis (HGF) is an uncommon slow-growing fibrous overgrowth characterized by connective tissue accumulation. It presents as an isolated feature or as a manifestation of any syndrome. Various syndromes associated with HGF are inherited by autosomal dominant/recessive/X-linked traits. Zimmermann–Laband syndrome (ZLS) is a rare, autosomal dominant inherited disease manifested with gingival fibromatosis (GF), nose and ears abnormalities, and hypoplastic/dysplastic nails or terminal phalanges of hand and feet. Although the pattern of inheritance was found to be both autosomal dominant and recessive traits, the molecular basis is still unclear. This report presents a possible case of ZLS-associated HGF in a 25-year-old female patient who presents with GF, hypertrichosis, and other syndrome-related features. Her father was similarly affected whereas her mother and sibling were asymptomatic. The patient and her family members were explained about the condition and surgical periodontal therapy was carried out for the patient to improve esthetics and was followed up regularly. Esthetics was significantly improved and no recurrence was noted at the end of 6 months.

Key words:

Gingival enlargement, hereditary gingival fibromatosis, syndromic gingival fibromatosis, Zimmermann–Laband syndrome

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INTRODUCTION

Hereditary gingival fibromatosis (HGF) is an uncommon gingival disease of genetic origin. It is a fibrous overgrowth that is benign in nature, grows slowly, and is characterized mainly by the accumulation of connective tissue fibroblasts. It presents either as an isolated feature or as a manifestation of any syndrome, covering the tooth to varying extents.[1] The 2017 classification of periodontal and peri-implant diseases and condition categorized gingival fibromatosis (GF) under nonbiofilm induced gingival diseases.[2] Nonsyndromic (isolated) HGF may be inherited by autosomal dominant trait whereas the syndromic form is characterized by autosomal dominant or autosomal recessive or even X-linked inheritance.[3] Various syndromes associated with HGF are GF with hypertrichosis, Zimmermann–Laband syndrome (ZLS), Ramon syndrome, Murray–Puretic–Drescher syndrome, Rutherford syndrome, Cross syndrome, Prune belly syndrome, Cantu syndrome, etc., ZLS is a rare heterogeneous congenital disorder affecting the facial areas and the extremities. Local or diffuse, nonhemorrhagic GF is the predominant manifestation followed by long and thin fingers and toes with hypoplastic or aplastic nails at birth. The secondary features are skeletal deformities of the maxillary arch, i.e., open bite, protruding jaws, macroglossia, high-arched palate, multiple unerupted teeth, abnormalities of soft cartilages of nose and ears, strabismus, hypertrichosis, intellectual deficiency, hepatosplenomegaly, hyperextensibility of joints, and mild hirsutism.[4] The pattern of inheritance was found in both autosomal dominant and recessive traits, but the molecular basis is still unclear. This case report presents a possible case of ZLS and its periodontal management.

CASE REPORT

A 25-year-old female patient reported with the chief complaint of enlarged gums of several years duration. A history revealed that the patient had enlarged gums since childhood and it slowly progressed to the present size. Family history revealed that she was not born out of

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Figure 1: (a) Extraoral appearance of the patient presenting with squint eye, bulbous nose, and thick lips; (b) Clinical appearance of the patient presenting with hypertrichosis and cutaneous pigmentation in the feet

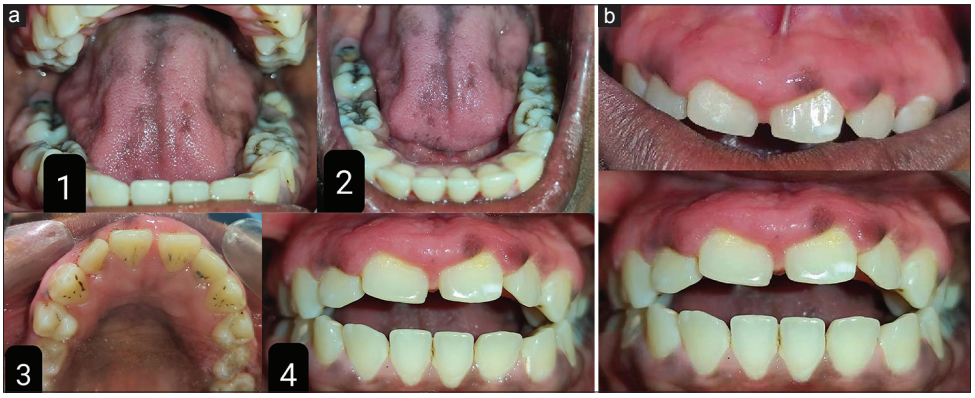


Figure 2: (a) Intraoral appearance of the patient presenting with (1) macroglossia, (2) multiple carious lesions, (3) high-arched palate, and (4) skeletal open bite; (b) Intraoral appearance of the patient presenting with Bokenkamp's grade III gingival enlargement



Figure 3: Orthopantomogram of the patient

consanguineous marriage and her father had similar gingival enlargement. Her mother and elder sister were apparently normal.

On extra oral examination, the patient had strabismus (squint eye), bulbous nose, prominent maxilla, and mandible with thick lips. The patient had hypertrichosis with terminal hairs in the legs, back region, and vellus hair in the face and other body parts. The patient also had cutaneous pigmentation in the feet [Figure 1a and b]. Intellectual and adaptive functioning of the patient was also impaired.

On intraoral examination, the patient had fair oral hygiene status with multiple carious lesions and also had macroglossia, high arched palate with skeletal open bite [Figure 2a]. Bokenkamp's grade III gingival enlargement was present in the buccal aspect of all maxillary teeth. The enlargement was diffuse, noninflammatory, and nonhemorrhagic involving marginal, attached, and interdental gingiva. Gingiva was pink in color with discrete areas of physiological melanin pigmentation. Margins were rounded and, on palpation, gingiva was non-tender and leathery in consistency. The surface texture was smooth with exaggerated stippling. The size was enlarged and the position was coronal to cementoenamel junction. Pseudopockets were present in the upper anterior region with no bleeding on probing [Figure 2b]. Orthopantomogram revealed no bone pathology [Figure 3]. Hematological and biochemical investigations were carried out. The investigatory findings were within normal limits, and the patient was systemically healthy. The patient's parents and siblings were examined on request. Oral examination of her father revealed poor oral hygiene status with similar gingival enlargement in the buccal and palatal/lingual aspect of posterior teeth of both the arches [Figure 4]. Her mother and sibling were found to be asymptomatic.

A genetic test on SOS-1 gene in 2p21-p22 chromosome was carried out to rule out nonsyndromic HGF. 5 ml of venous blood was collected in ethylenediaminetetraacetic acid-coated tubes. Genomic DNA was extracted and stored at 20°C, and the amplification process was carried out with an initial denaturation at 95°C for 5 min. The process was repeated for 35 cycles at 95°C for 45 s, 61°C for 45 s, and 72°C for 45 s with a final extension of 72°C for 5 min. The amplicons were further purified and directionally sequenced using BigDye terminator sequencing and ABI genetic analyzer (Outsourced - Eurofins Genomics India Pvt. Ltd). The sequence obtained was compared with the reference genome in the database. Multiple sequence alignment using the Clustal Omega software did not show



Figure 4: Intraoral appearance of the patient's father showing gingival fibromatosis

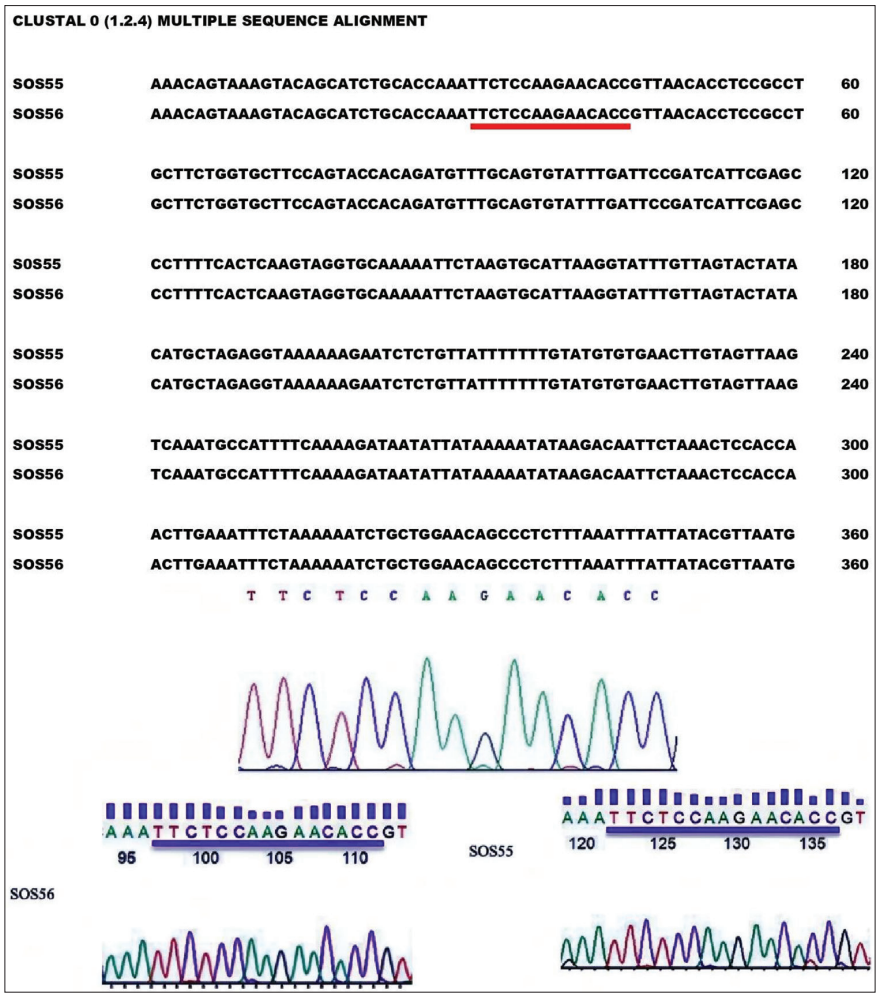


Figure 5: Results of DNA sequencing of the patient and her father showing no significant mutation in SOS-1 gene



Figure 6: Intraoperative views: (1) Showing placement of surgical guide (2) Showing external bevel gingivectomy (3) Showing placement of periodontal dresssing

Table 1: Syndromes associated with hereditary gingival fibromatosis

Syndrome	Inheritance	Features
GF with hypertrichosis	AD	GF, hypertrichosis, mental retardation, muscular hypotonia
GF with distinctive facies	AR	GF, macrocephaly, bushy eyebrows, synophrys, hypertelorism, down-slanted palpebral fissures, flat nasal bridge, hypoplastic nares, cupid-bow mouth, high arched palate
Jones syndrome	AD	GF, progressive sensory, neural hearing loss
Zimmermann–Laband syndrome	AD	GF–hypertrophy, absence/dysplasia of the terminal phalanxes or nails of the hands or feet and thick lips, bulbous soft nose, thick floppy ears, mental retardation, hepatosplenomegaly, hypertrichosis, hyperextensibility of the joints, ocular symptoms
Ramon syndrome	AR	GF, cherubism, seizures, mental deficiency, hypertrichosis, stunted growth, juvenile rheumatoid arthritis
Juvenile hyaline fibromatosis (Murray-Puretic-Drescher)	AR	GF, multiple subcutaneous tumors, sclerodermiform atrophy, osteolytic and osteoblastic skeletal lesions, recurrent suppurative infections, painful flexural joint contractures, osteolysis of terminal phalanges, stunted growth/early death
Systemic infantile hyalinosis	AR	Gingival hypertrophy, thickened skin/focal skin nodularity, joint contractures/osteoporosis, diarrhea/failure to thrive, recurrent infections/death (infancy)
Rutherford syndrome	AD	Gingival hypertrophy, corneal opacity, mental retardation, failure of tooth eruption, aggressive behavior
Cross syndrome	AR	Hypopigmentation/silver grey hair color, microphthalmia with cloudy corneas, mental retardation/spasticity, athetoid movements/growth retardation
Prune belly	Unclear	Absence of abdominal muscles, abnormalities of urinary tract, cryptorchidism, facial dimorphism
Gangliosidosis	X-linked	Gingival hypertrophy, macroglossia, coarse face/micrognathia, loose skin/inguinal hernia, delayed growth/hepatosplenomegaly, neonatal hypotonia, delayed motor development
Borrone dermato-cardio-skeletal syndrome	AR, X-linked	Gingival hypertrophy/coarse face, late eruption of teeth/loss of teeth, thick skin/acne conglobata, osteolysis/large joint flexion contractures, short stature/brachydactyly/congestive heart failure, inguinal hernia
Mannosidosis	AR	Gingival hypertrophy/macroglossia, coarse features/prognathism, thick eyebrows/low anterior hairline, deafness/lens opacities, hepatosplenomegaly, recurrent respiratory tract infections, muscular hypotonia/mental retardation
I-cell disease (mucolipidosis)	AR	Coarse facial features, normal head circumference relative to body size, puffy eyelids with slight exophthalmia, excessive prominence of the epicanthic folds, depressed nasal bridge, full cheeks exhibiting multiple fine telangiectasia, incompetent lips, gingival and alveolar enlargement with buried teeth, thick tongue
Donohue syndrome–leprechaunism	AR	Hirsutism, acanthosis nigricans, large mouth, thick lips, gingival hypertrophy, paucity of lymphatic tissue, hepatic cholestasis, and fibrosis, large hands, feet, retarded bone age, reduced muscle mass, and distended abdomen
Cantu syndrome	Possibly AR	Congenital hypertrichosis, osteochondrodysplasia, cardiomegaly, mental retardation, macrocephaly, enlarged sella turcica, prominent mouth, narrow shoulders-thorax, broad ribs, platyspondyly, and coxa valga

AR – Autosomal recessive; GF – Gingival fibromatosis; AD – Autosomal dominant

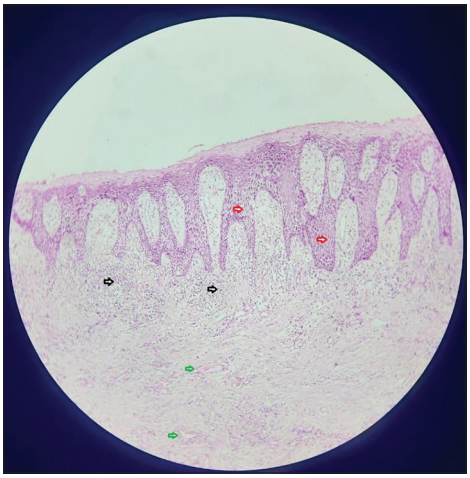


Figure 7: Photomicrograph showing hyperplastic epithelium (red arrows) and dense fibrous connective tissue exhibiting hyalinization of collagen (green arrows) with focal areas of inflammatory cells (black arrows)

any prominent variations between the samples [Figure 5]. Hence, nonsyndromic HGF was ruled out. Thus, after ruling out the possibility of a nonsyndromic association, the case was provisionally diagnosed as syndromic HGF. Based on the history, clinical findings and investigations, the case was finally diagnosed as ZLS-associated HGF.



Figure 8: Postoperative views – (1) 1 week; (2) 6 months

Hopeless teeth were extracted, and phase-I periodontal therapy was carried out. Gingivectomy was planned to improve esthetics. External bevel gingivectomy with a surgical guide was done under local anesthesia at 2 weeks' interval in the areas with GF. The periodontal dressing was placed and 0.12% chlorhexidine mouth rinse was advised for 2 weeks, postsurgically [Figure 6]. The excised gingival tissues were sent for histopathological examination. Histopathology revealed the presence of hyperplastic parakeratinized squamous epithelium of variable thickness and bundles of dense collagenous fibrous connective tissue exhibiting hyalinization of collagen with focal areas of inflammatory cells [Figure 7]. Histopathological diagnosis was correlated with the clinical diagnosis and was found to be

compatible. The patient was followed up periodically for plaque control to prevent recurrence. Esthetics in terms of gingival appearance improved significantly and no recurrence was noted at the end of the 6th month, postoperatively [Figure 8].

DISCUSSION

GF is a slow-growing fibrous overgrowth of gingival connective tissue, caused by various factors such as drugs, hormonal changes, and genetics. It can also present as a manifestation of systemic diseases such as leukemia, lymphoma, Wegener's granulomatosis, sarcoidosis, and oral granulomatosis. The genetically induced GF is of two forms – syndromic and nonsyndromic. Nonsyndromic HGF might be inherited by a single-gene mutation whereas multiple gene mutations or gene dosage effect might result in syndromic HGF.[5] The first case of HGF reported by William H Goddard was found to be nonsyndromic with the autosomal dominant trait, where the mutation was noted in son of sevenless-1 gene in locus 2p21-p22.[6] Chromosomes 2p13-p16, 4q21, and 5q13-22 are also genetically linked to the autosomal dominant trait of HGF. The inheritance pattern of syndromic HGF might be autosomal dominant or recessive or even X-linked trait. Various syndromes associated with GF are listed in Table 1.[1,3,7,8]

In the present case report, a 25-year-old female patient was diagnosed as a possible case of ZLS based on the familial history, clinical features, and hematological, biochemical, and histopathological investigations. ZLS is a rare autosomal dominant congenital disorder affecting facial areas and extremities. The mutation in the CACNA2D3 gene, LRTM1gene, WTN5A gene, and ERC2 gene located in chromosome 3p14.3 is responsible for ZLS.[9] Stefanova *et al*. have assessed the 3p breakpoint location and have found mutation in CACNA2D3 gene and WTN5A gene in a mother and daughter with ZLS.[10] Hoogendijk *et al*. have documented a chromosomal insertion in chromosome 8 (12;8) (p11.2; - q11.2q24.3) in a patient with ZLS and stated that mutation in chromosome 8 is the prime etiology for the syndrome.[11] Shirian *et al*. have detected mutations at 3q13.13, 1q25.2, 1p12, and also in 19q12 in a patient with typical clinical features of ZLS.[12] Guglielmi *et al*.[13] have identified mutation in KCNH1 gene and Galal *et al*.[14] have detected mutation in ATP6V1B2 gene, respectively. Even though various studies have reported different inheritance patterns for ZLS, the molecular basis is still not clear.

The clinical presentation of HGF may vary, exhibiting minimal involvement characterized by enlargement of maxillary tuberosity and mandibular molar gingiva, to severe involvement encompassing both maxillary and mandibular gingiva covering the crowns of the teeth completely. In the present case report, the involvement was said to be moderate as the GF has involved maxillary gingiva alone covering 2/3rd the crowns of the teeth. However, the patient was presented with the manifestations of ZLS without phalangeal involvement. Castori *et al*. have reported that gingival hypertrophy with and without phalangeal involvement has been considered explicit characteristic of ZLS.[15] Shah and Gupta have reported a case of a 3.5-year-old child with ZLS associated with bilateral cataract.[16] Lin *et al*. documented a case of ZLS with thymic hyperplasia, ventricular hypertrophy, and aortic dilation.[17] Since the patient reported after puberty, thymic hyperplasia was not evaluated directly as thymus undergoes genetic involution at puberty. However, thymic hyperplasia is usually associated with myasthenia gravis and hyperthyroidism in most of the cases. Hence, symptoms of myasthenia gravis such as shortness of breath, difficulty in swallowing, blurred or double vision, weakness of eye muscles, impaired speech, and ptosis of the eyelid were evaluated. No significant symptoms were noted and hence ruled out. To rule out hyperthyroidism, thyroid function tests were performed. Thyroid levels were normal and hence, ruled out. Ventricular hypertrophy and aortic dilation were ruled out as no significant clinical symptoms such as chest pain, palpitations, shortness of breath and heart murmurs were noted. Since clinical interpretations were negative, no radiographic or advanced imaging interventions were carried out further.

The histologic appearance of HGF is usually characterized by parakeratinized, squamous hyperplastic epithelium with elongated rete pegs and a fibrous, acellular connective tissue with fibroblast cells. Calcified particles, amyloid deposits, and odontogenic epithelial islands are unusual findings.[18] The genetic mutation increases the production of β-1 transforming growth factor (TGF-β1), which enhances the transdifferentiation of gingival myofibroblasts resulting in GF. There are histopathological and cell culture reports in the literature which documented the presence of myofibroblasts. The recent research stated that the interferon-γ could attenuate the transdifferentiated myofibroblast-induced extracellular matrix accumulation and thus, might be helpful in the prevention of HGF.[19]

Guglielmi *et al*. have documented an increased inflammatory state proteins-defensins alpha 1 and 3 in salivary proteome of a 3-year-old child with ZLS compared to healthy subjects of the same sex and age and also noted the presence of S100A7. Thus, proteomic analysis serves as an adjunctive method of diagnosis of rare diseases with differentially expressed proteins and peptides.[13]

The periodontal management can be done surgically to improve the esthetics by restoring the gingival contours. The surgical management includes external or internal bevel gingivectomy, and apically displaced flap which can be done with a scalpel or cryosurgery and laser, but the predictability of long-term results is still unclear. It has been documented in various literature that the plaque accumulation mediates the recurrence rate of fibromatosis as the recurrence is higher in areas with severe plaque accumulation and minimal or delayed in patients with good oral hygiene. Hence, the treated patients should be recalled and reviewed at regular intervals. Professional cleaning and oral hygiene instructions should be reinforced to reduce the recurrence rate. In this case report, quadrant-wise external bevel gingivectomy was done at regular intervals, and the patient was followed up periodically. A significant improvement in the esthetics and no recurrence was noted at the end of 6 months.

CONCLUSION

HGF can present as an isolated feature or as one of the features of multisystem syndrome, which is associated with an altered social, emotional, and functional characteristics of

an individual. ZLS is not a life-threatening disease but usually presents with several medical, dental, and psychological problems. Hence, a multidisciplinary approach, involving geneticist, periodontist, pathologist, biochemist, and psychologist, is required for proper diagnosis and appropriate treatment of the disease. More studies should be focused on proteomics to identify molecular markers for early diagnosis and correlation of dental and genetic phenotypes. TGF-β1 has found to be primarily associated with GF, but evidence supporting counteraction of TGF-β1 are still lacking and needs to be explored. Thus, future studies investigating the suppressing action of interferon against TGF-β1 should be evolved to prevent the development of HGF.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Conflicts of interest

There are no conflicts of interest.

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